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Course: Foundation Of Python Programming
Assignment06
GitHub Link: https://github.com/romulojrreyesrr/IntroToProg-Python-Mod06

**Assignment06: Student Course Registration using Static method function and Class IO features**

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## Introduction

This Python program allows a user to register students for courses, display current enrollment data, save the data to a file, and exit the program.

The program uses:

- Constants
- Variables
- Classes
- Static methods
- JSON file processing
- Structured error handling

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## How the Program Works

When the program starts, it automatically reads data from the file **Enrollments.json** into a list called students.

The user is shown a menu with four options:

1. Register a student
2. Show current data
3. Save data to a file
4. Exit the program

```python
# ------------------ Main Program ------------------ #

# Load existing data at startup
FileProcessor.read_data_from_file(FILE_NAME, students)

while True:
    IO.output_menu(MENU)
    menu_choice = IO.input_menu_choice()

    if menu_choice == "1":
        IO.input_student_data(students)

    elif menu_choice == "2":
        IO.output_student_courses(students)

    elif menu_choice == "3":
        FileProcessor.write_data_to_file(FILE_NAME, students)

    elif menu_choice == "4":
        print("Program ended.")
        break

    else:
        IO.output_error_messages("Please select a valid option (1-4).")
```

```
---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course
    2. Show current data
    3. Save data to a file
    4. Exit the program
----------------------------------------

Enter your menu choice [1-4]:
```

The program runs in a loop until the user selects option 4.

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## IO Class

Handles all input and output tasks such as:

- Displaying the menu
- Getting user input
- Showing student data
- Displaying error messages

```python
class IO:
    """Performs Input and Output tasks for the Course Registration Program."""

    @staticmethod
    def output_error_messages(message: str, error: Exception = None):
        """Displays structured error messages."""
        print("ERROR:", message)
        if error:
            print("Technical Details:", error)

    @staticmethod
    def output_menu(menu: str):
        """Displays the program menu."""
        print(menu)

    @staticmethod
    def input_menu_choice():
        """Gets the user's menu choice."""
        return input("Enter your menu choice [1-4]: ").strip()

    @staticmethod  2 usages
    def output_student_courses(student_data: list):
        """Displays student enrollment data."""
        print("\nCurrent Enrollment Data:")
        for row in student_data:
            print(f"{row['FirstName']},{row['LastName']},{row['CourseName']}")
        print()
```

```
ERROR: File not found. Starting with empty list.
Technical Details: [Errno 2] No such file or directory: 'Enrollments.json'

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course
    2. Show current data
    3. Save data to a file
    4. Exit the program
----------------------------------------

Enter your menu choice [1-4]:
```

```
Enter your menu choice [1-4]: 1
Enter student's first name: Rche
Enter student's last name: Boo
Enter course name: Python 100
Student registration added.


---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course
    2. Show current data
    3. Save data to a file
    4. Exit the program
----------------------------------------

Enter your menu choice [1-4]: 2

Current Enrollment Data:
Rche,Boo,Python 100
```

## FileProcessor Class

Handles file operations such as:

- Reading data from the JSON file
- Writing data to the JSON file

```python
@staticmethod
def write_data_to_file(file_name: str, student_data: list):
    """Writes student enrollment data to a JSON file."""
    try:
        file = open(file_name, "w")
        json.dump(student_data, file, indent=2)
        file.close()
        print("Data successfully saved to file.")
        IO.output_student_courses(student_data)
    except Exception as e:
        IO.output_error_messages( message: "\033[91mWarning:Error writing data to file.\033[0m", e)
```

```python
class FileProcessor:
    """Processes data to and from a JSON file."""

    @staticmethod
    def read_data_from_file(file_name: str, student_data: list):
        """Reads student enrollment data from a JSON file."""
        try:
            file = open(file_name, "r")
            student_data.clear()
            student_data.extend(json.load(file))
            file.close()
        except FileNotFoundError as e:
            IO.output_error_messages( message: "File not found. Starting with empty list.", e)
        except json.JSONDecodeError as e:
            IO.output_error_messages( message: "File contains invalid JSON.", e)
        except Exception as e:
            IO.output_error_messages( message: "Unexpected error while reading file.", e)

    @staticmethod
    def write_data_to_file(file_name: str, student_data: list):
        """Writes student enrollment data to a JSON file."""
        try:
            file = open(file_name, "w")
            json.dump(student_data, file, indent=4)
            file.close()
            print("Data successfully saved to file.")
            IO.output_student_courses(student_data)
        except Exception as e:
            IO.output_error_messages( message: "Error writing data to file.", e)
```

```
---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course
    2. Show current data
    3. Save data to a file
    4. Exit the program
----------------------------------------

Enter your menu choice [1-4]: 3
Data successfully saved to file.

Current Enrollment Data:
Rche,Boo,Python 100
Czeon,Roche,Python 100
Romulo,Reyes,Python 100
```

```
Enrollments.json
[
  {
    "FirstName": "Rche",
    "LastName": "Boo",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Czeon",
    "LastName": "Roche",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Romulo",
    "LastName": "Reyes",
    "CourseName": "Python 100"
  }
]
```

---

## Data Storage

Student data is stored in a list of dictionaries. Each student record includes:

- FirstName
- LastName
- CourseName

```python
@staticmethod
def write_data_to_file(file_name: str, student_data: list):
    """Writes student enrollment data to a JSON file."""
    try:
        file = open(file_name, "w")
        json.dump(student_data, file, indent=2)
        file.close()
        print("Data successfully saved to file.")
        IO.output_student_courses(student_data)
    except Exception as e:
        IO.output_error_messages( message: "\033[91mWarning:Error writing data to file.\033[0m", e)
```

```
---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course
    2. Show current data
    3. Save data to a file
    4. Exit the program
----------------------------------------

Enter your menu choice [1-4]: 3
Data successfully saved to file.

Current Enrollment Data:
Rche,Boo,Python 100
Czeon,Roche,Python 100
```

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## Error Handling

The program includes error handling for:

- Invalid first name input
- Invalid last name input
- File not found errors
- Invalid JSON file format
- File writing errors
- Invalid menu choices

```python
@staticmethod
def input_student_data(student_data: list):
    """Collects student enrollment data from the user."""
    try:
        first_name = input("Enter student's first name: ").strip()
        if not first_name.isalpha():
            raise ValueError("\033[1;33mFirst name must contain only letters\033[0m.")

    except Exception as e:
        IO.output_error_messages( message: "\033[1;33mInvalid first name.\033[0m", e)
        return

    try:
        last_name = input("Enter student's last name: ").strip()
        if not last_name.isalpha():
            raise ValueError("\033[1;33mLast name must contain only letters.\033[0m")

    except Exception as e:
        IO.output_error_messages( message: "\033[1;33mInvalid last name.\033[0m", e)
        return
```

```
Enter your menu choice [1-4]: 1
Enter student's first name: Romulo 100
ERROR: Invalid first name.
Technical Details: First name must contain only letters.

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course
    2. Show current data
    3. Save data to a file
    4. Exit the program
----------------------------------------
```

---

## Conclusion

This program successfully allows users to manage student course registrations. It demonstrates the use of classes IO, static method functions, file handling, and structured error handling in Python.